

Overview

We developed an Object Detection and Actuation System on a mobile-centric platform using YOLO and a robot-mounted camera. The system detects objects in real-time and triggers actions based on the identified objects.



Impact of System

- Deployment of a simple lightweight YOLOv5s detector on resource-constrained hardware (RPI5)
- Real-time and closed-loop perception-action pipeline for object detection and physical response
- Quantitative evaluation using Pascal VOC-style test samples
- Analysis of detection confidence, bounding box dimensions, and per-image metrics
- Visual diagnostics: confidence histograms, detection maps, and IoU-based insights

*Want to see it in action?
Just scan and explore!*



Mathematical Analysis

Confidence Score

$$C = P(\text{object}) \cdot \max_{c \in \text{classes}} P(c | \text{object})$$

$$\text{IoU} = \frac{\text{Area}(B_p \cap B_{gt})}{\text{Area}(B_p \cup B_{gt})}$$

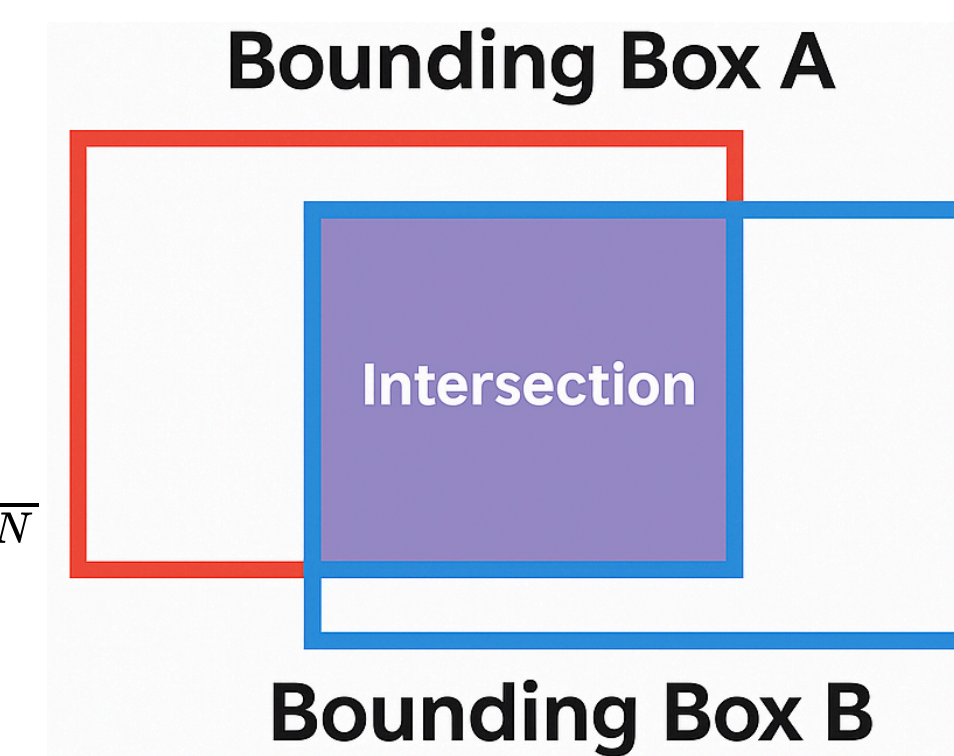
Precision and Recall

$$\text{Precision} = \frac{TP}{TP+FP} \quad \text{Recall} = \frac{TP}{TP+FN}$$

Yolo Loss function

$$\mathcal{L}_{\text{total}} = \int_{\Omega} \left[\lambda_{\text{box}} \cdot \mathcal{L}_{\text{CIoU}}(b, \hat{b}) + \lambda_{\text{obj}} \cdot \mathcal{L}_{\text{BCE}}(p_{\text{obj}}, \hat{p}_{\text{obj}}) + \lambda_{\text{cls}} \cdot \mathcal{L}_{\text{BCE}}(p_{\text{cls}}, \hat{p}_{\text{cls}}) \right] d\Omega$$

Intersection over Union

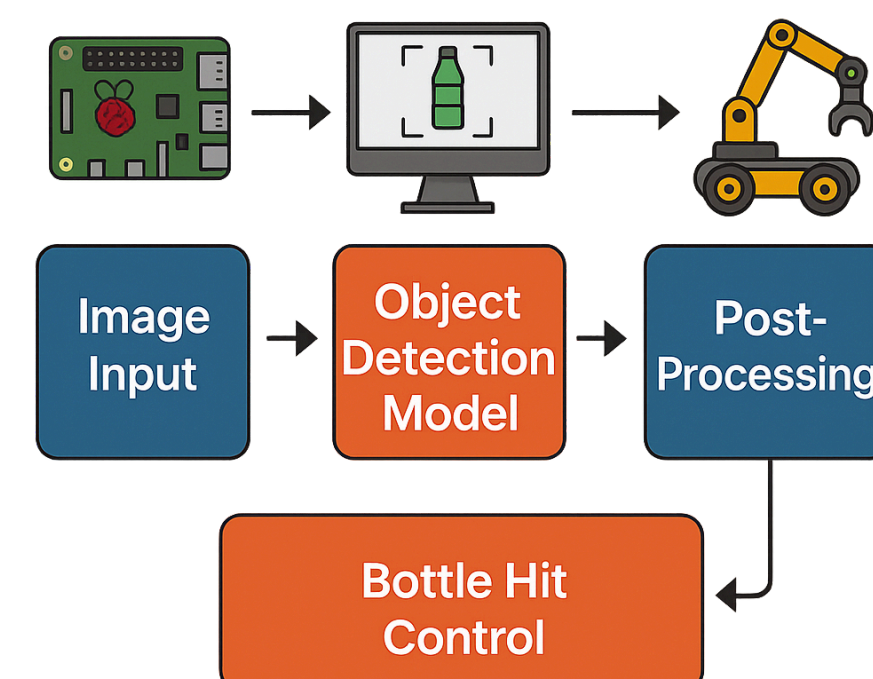


System Overview

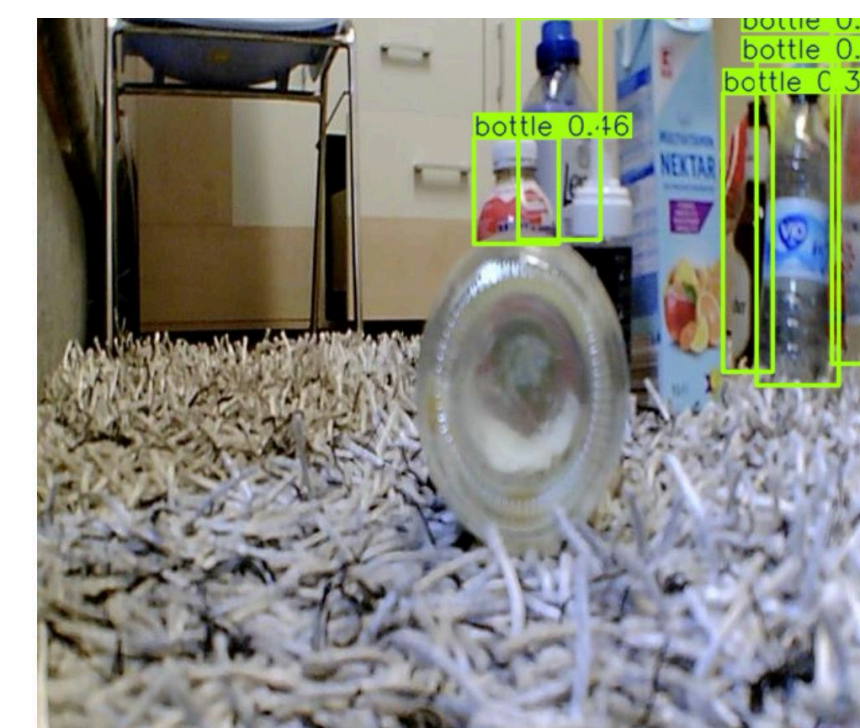
Hardware Requirements

- Raspberry Pi 5 as main CPU
- CSI camera module for image acquisition
- DC motor with driver for mobility
- Servo-controlled arm to strike detected bottles
- 12V power supply with 5V converter for Raspberry Pi

Software Pipeline



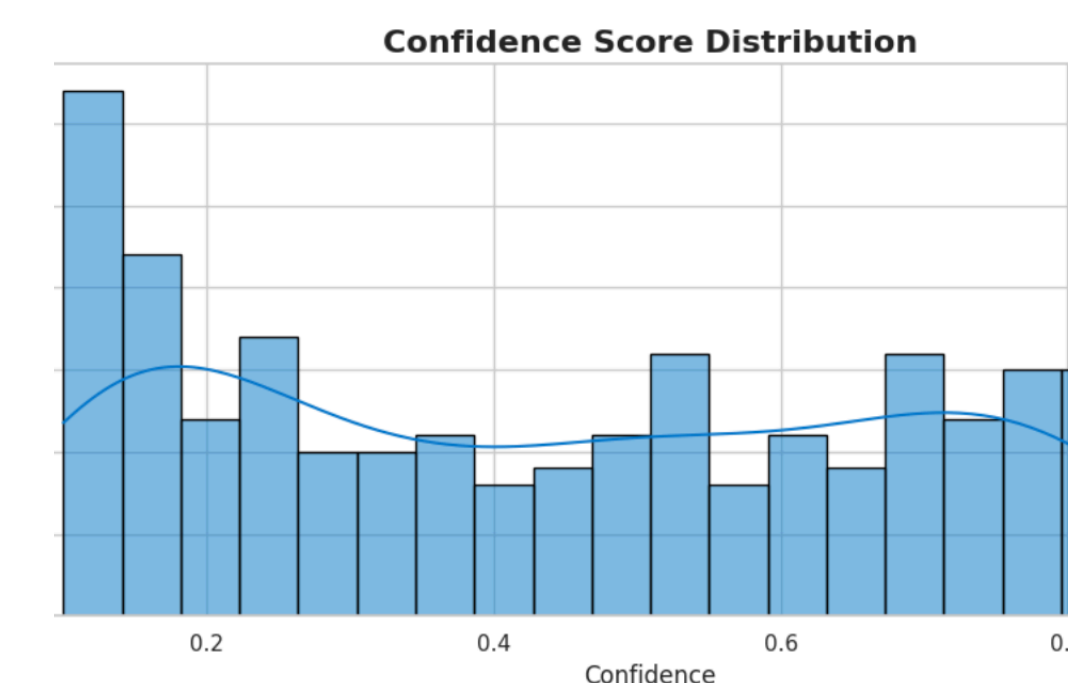
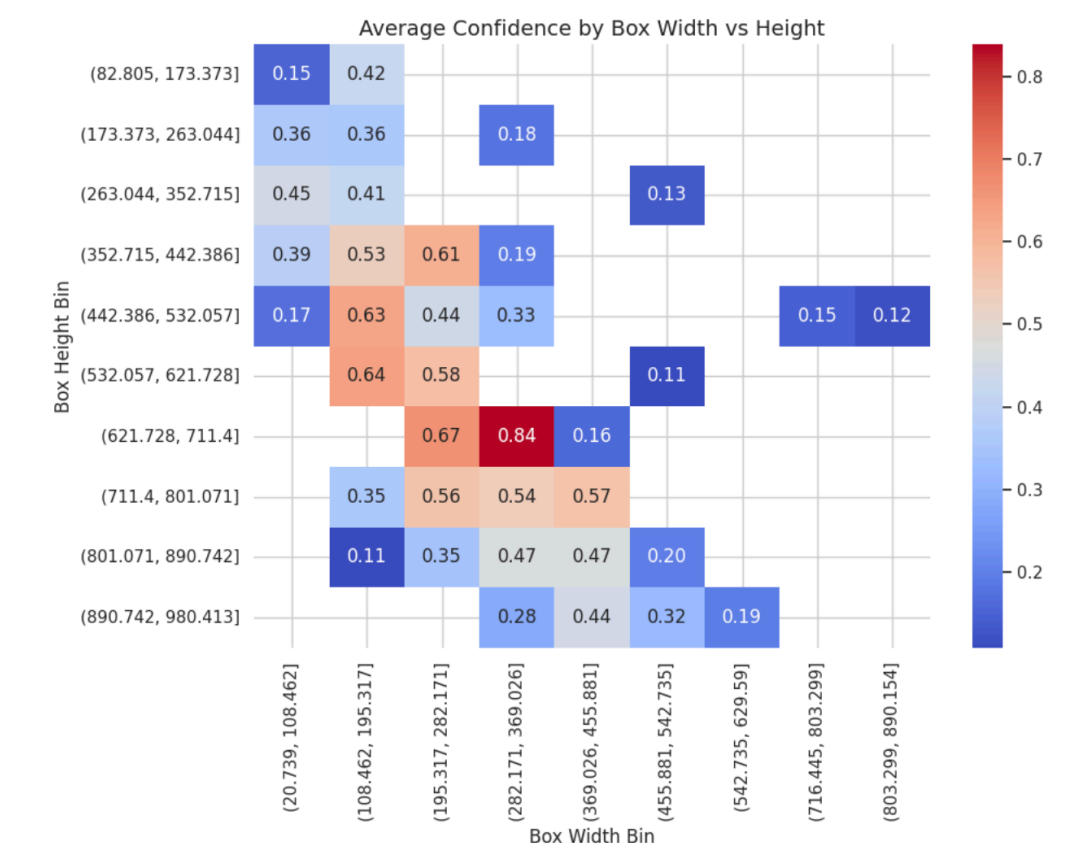
Post detection response time is measured and actions are logged.



Experimental Results

Heatmap: Confidence-Width × Height

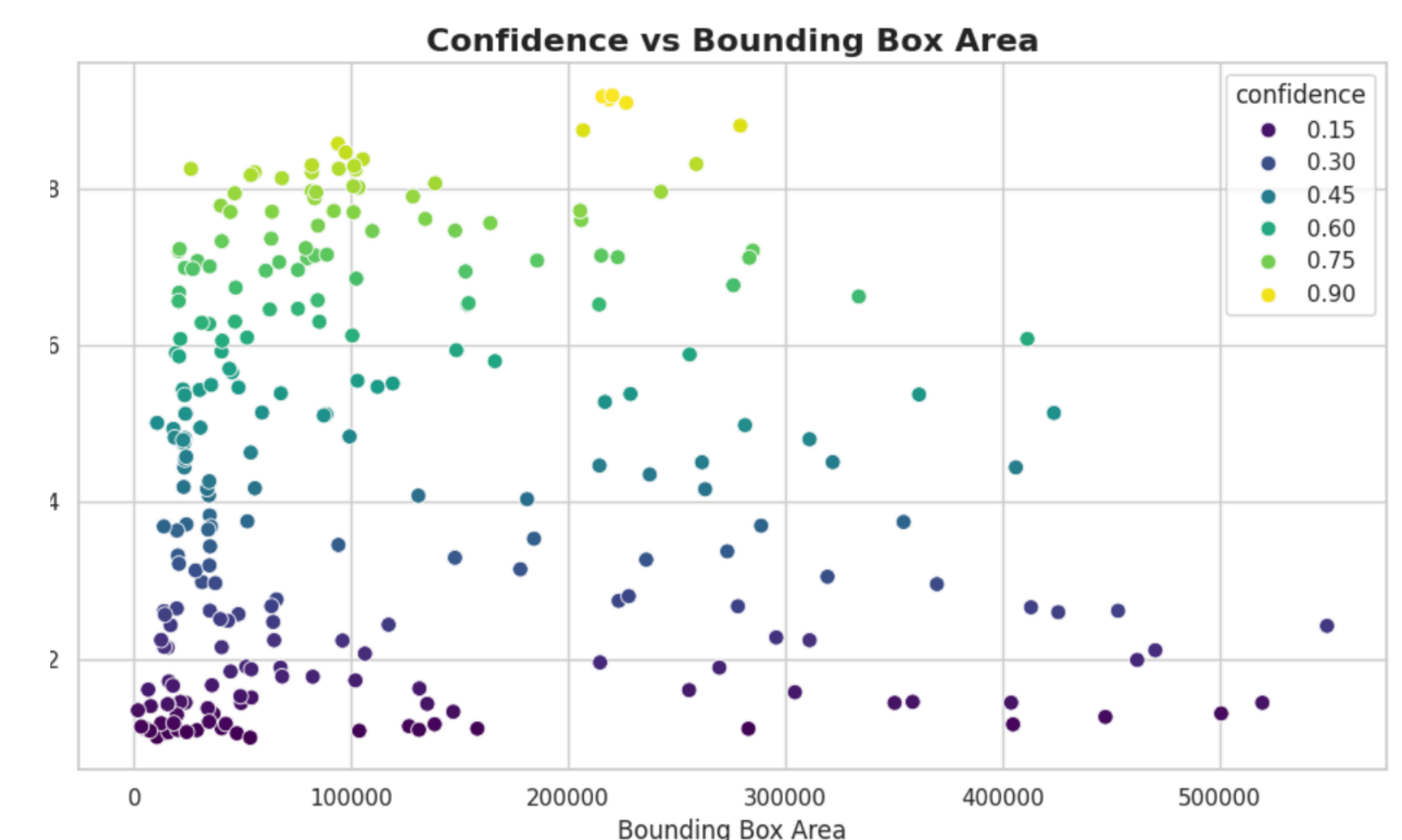
The heatmap shows peak confidence (0.84) for 300–700 px boxes, with sharp drops for distorted aspect ratios.



Confidence Score Distribution

The bimodal histogram shows false positives below 0.2 and reliable detections at 0.6–0.8, validating the 0.5 threshold.

Confidence vs Bounding Box Area



YOLOv5s shows highest confidence for medium-sized bottles; very small or large boxes score lower.

References

- 1] Bochkovskiy, A., Wang, C.Y., & Liao, H.Y.M. (2020). YOLOv4: Optimal Speed and Accuracy of Object Detection. arXiv:2004.10934.
- 2] <https://github.com/ultralytics/yolov5> (Realtime Instance Segmentation)
- 3] Everingham, M., et al. (2010). The Pascal Visual Object Classes (VOC) Challenge. IJCV, 88(2), 303–338.